

MSE of Exponential Rate Estimators under Right Censoring: a simulation study

2AMS11 Survival analysis for Data-Science

Week 2

Introduction

In this group assignment (3 to 4 students), you will conduct a simulation study and gain hands-on experience with MSE calculations. This exercise also serves as preparation for the larger assignment that follows.

You may use large language models (LLMs), but any use of AI tools must be explicitly documented. Guidelines for documenting AI use are provided at the end of this document. In addition, you must clearly state the contribution of each group member; instructions are also provided at the end of this document.

Your report should be no longer than **one page**, excluding the contribution statement and AI-use declaration. The grading criteria can be found at the end of this document.

Learning Objectives

- Design and implement a simulation study.
- Assess estimator performance using MSE.
- Communicate results clearly and critically.

Goal of Simulation Study

Compare two estimators of the exponential rate λ under right censoring:

- Naive estimator that ignores censoring: $\hat{\lambda}_{\text{naive}} = 1/\bar{Y}$ where $Y_i = \min(T_i, C_i)$.
- Correct MLE under right censoring: $\hat{\lambda}_{\text{mle}} = r / \sum_{i=1}^n Y_i$, where $r = \sum_i \delta_i$ and $\Delta_i = \mathbb{I}\{T_i \leq C_i\}$.

Estimate the mean squared error (MSE) of each estimator via Monte Carlo for multiple censoring proportions.

Model

True event times: $T_i \stackrel{\text{iid}}{\sim} \text{Exponential}(\lambda)$ for $i = 1, \dots, n$. Right-censoring times: C_i generated independently (use e.g. $\text{Uniform}(0, c_{\max})$ or Exponential with rate ρ) chosen so that the realized censoring proportion approximates a target p_c .

Observed data: $Y_i = \min(T_i, C_i)$ and $\Delta_i = \mathbb{I}\{T_i \leq C_i\}$.

Estimators

Given observed (y_i, δ_i) for $i = 1 \cdots n$:

$$\hat{\lambda}_{\text{naive}} = \frac{1}{\frac{1}{n} \sum_{i=1}^n y_i}$$

(treats all y_i as complete).

$$\hat{\lambda}_{\text{mle}} = \frac{r}{\sum_{i=1}^n y_i} \quad \text{where } r = \sum_{i=1}^n \delta_i.$$

(If $r = 0$ in a simulation replicate, record NA and handle appropriately.)

MSE definition (for each estimator):

$$\text{MSE}(\hat{\lambda}) = \mathbb{E}[(\hat{\lambda} - \lambda)^2],$$

estimated by Monte Carlo as the empirical mean of squared errors across replicates.

Simulation procedure

For each target censoring proportion $p_c \in 0\%, 10\%, 50\%$:

1. Choose sample size n and true rate λ (suggest $n = 200$ or 500 , $\lambda = 0.05$).
2. Choose a censoring distribution family (e.g., $C \sim \text{Uniform}(0, c_{\max})$). Calibrate $c_{\max}$ so that empirical censoring proportion $\approx p_c$ (you may tune by trial; a short calibration run of 1000 simulated T and varying $c_{\max}$ helps).
3. Repeat R times (suggest $R = 2000$):
 - (a) Simulate $T_i \sim \text{Exp}(\lambda)$, and C_i from the chosen censoring distribution.
 - (b) Compute $y_i = \min(t_i, c_i)$ and $\delta_i = \mathbb{I}\{t_i \leq c_i\}$.
 - (c) Compute $\hat{\lambda}_{\text{naive}}$ and $\hat{\lambda}_{\text{mle}}$ (if $r = 0$, mark replicate as NA for mle).
4. Compute empirical MSE for each estimator using non-NA replicates (for mle exclude NA or report separately).

Reporting

For each p_c report:

- average censoring proportion (observed across replicates),
- empirical bias and variance for each estimator,
- MSE for each estimator,
- proportion of replicates with $r = 0$ (if any).

Discuss how ignoring censoring affects bias and variance as censoring increases, and why the MLE is preferred.

Deliverables

- Report including AI and contribution statements
- Code

Length of report should be maximal 1 page excluding the AI and contribution statements.

Use of Artificial Intelligence Tools

The use of LLMs is permitted. However, full transparency is required.

For every use of an LLM, report:

- Date of use.
- Name of the tool.
- Exact prompt(s) used.
- Summary of the output received.
- How the output was used.
- Whether modifications were made.

Students remain fully responsible for the correctness and integrity of all submitted work. Failure to disclose AI use may be considered academic misconduct.

Contribution Statement

Each report must end with a contribution statement.

For each group member, describe contributions to:

- Problem formulation.
- Statistical design.
- Programming.
- Analysis.
- Writing and editing.

Example:

Student A developed the practical application and simulation design.

Student B implemented the simulation code and generated figures.

Student C conducted the statistical analyses and wrote the conclusions.

All group members must confirm that the contribution statement is accurate.

Grading Rubric

- Code: 1
- Correct implementation: 2
- Interpretation, discussion: 1
- Appropriate use of AI: 0.5
- Presentation of results: 0.5